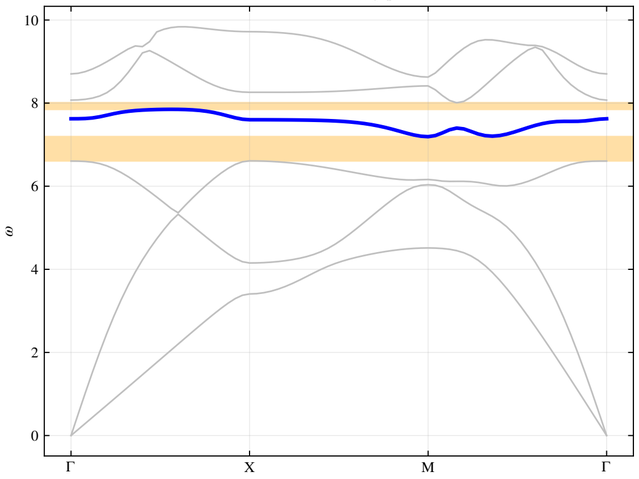
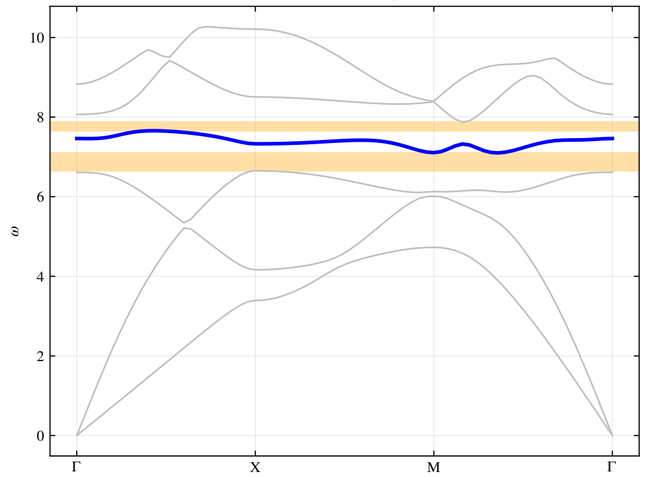
Flat-band metamaterial: jointly refining mesh n, orientation supports k_s, and tow pitch d

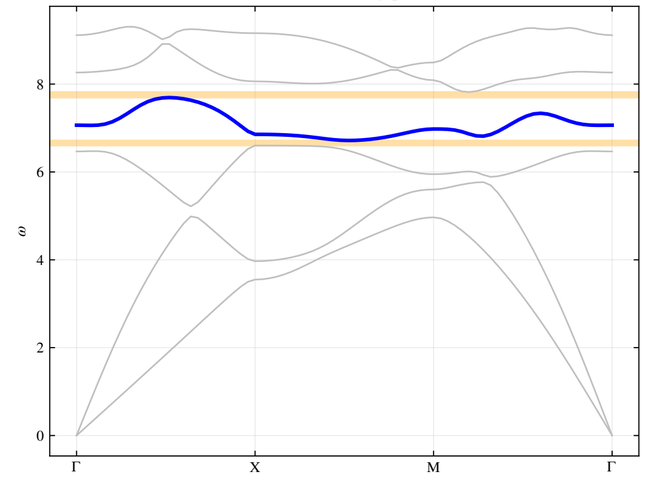
(a) C1: n=32, k_s=4 (M=16), d=0.167
band 3: 2.87 → 0.66 (4.4×), gaps +0.16/+0.58



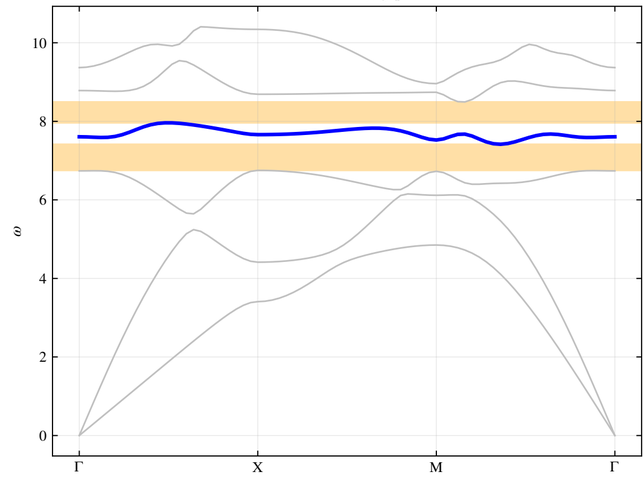
(b) C2: n=32, k_s=6 (M=36), d=0.111
band 3: 2.52 → 0.56 (4.5×), gaps +0.22/+0.45



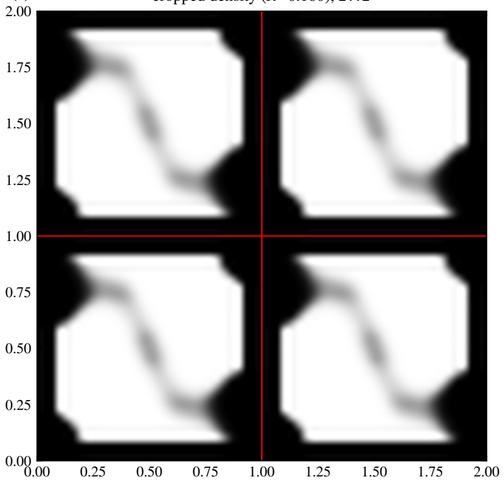
(c) C3: n=32, k_s=8 (M=64), d=0.083
band 3: 2.06 → 0.98 (2.1×), gaps +0.13/+0.12



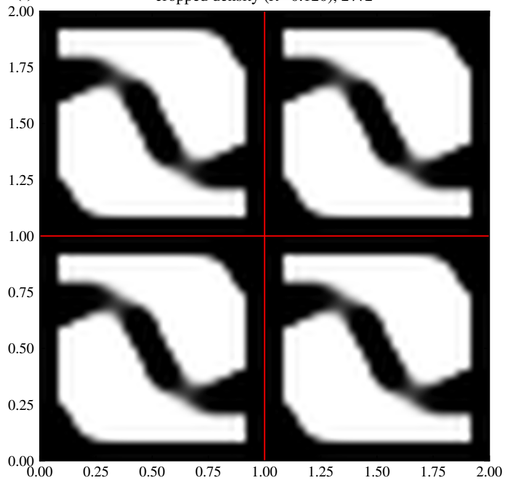
(d) C3d: n=32, k_s=8 (M=64), d=0.083
band 3: 2.06 → 0.55 (3.8×), gaps +0.53/+0.67



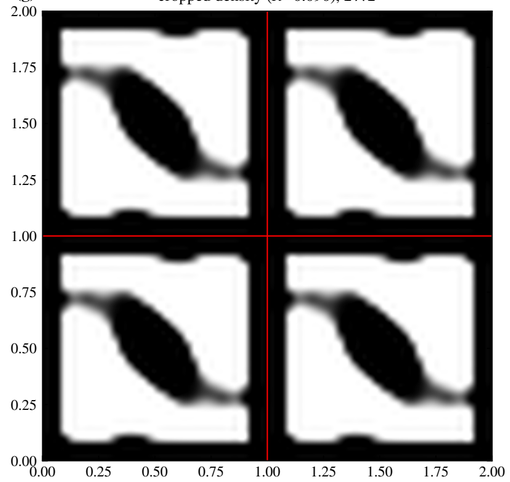
(e) cropped density (R=0.160), 2×2



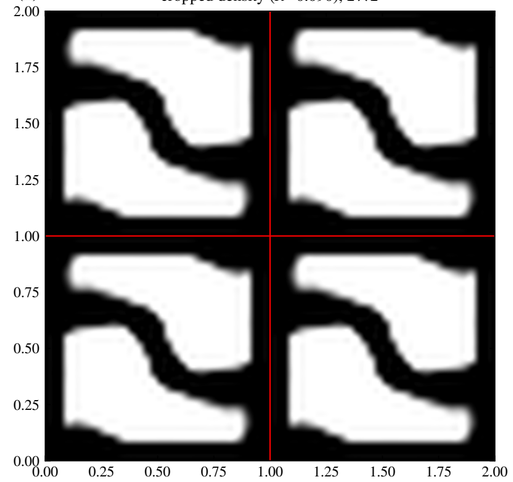
(f) cropped density (R=0.120), 2×2



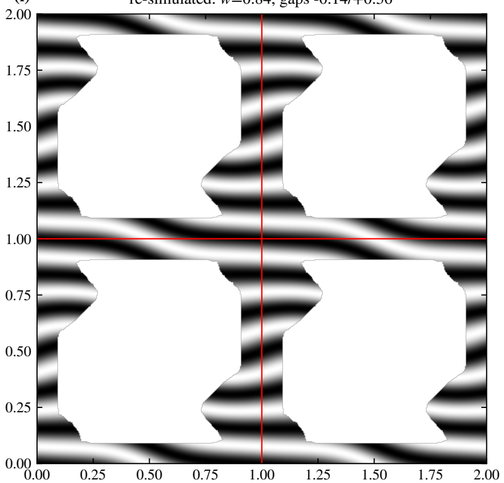
(g) cropped density (R=0.090), 2×2



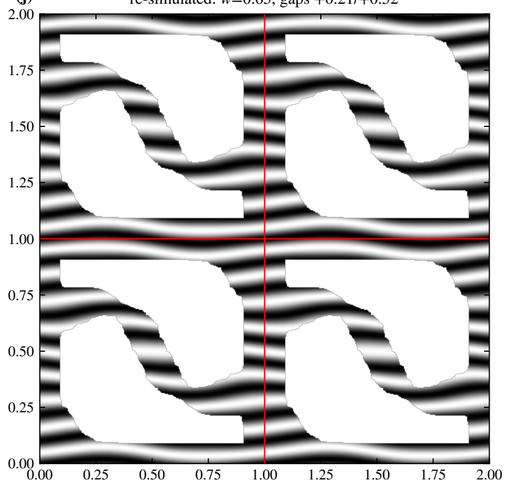
(h) cropped density (R=0.090), 2×2



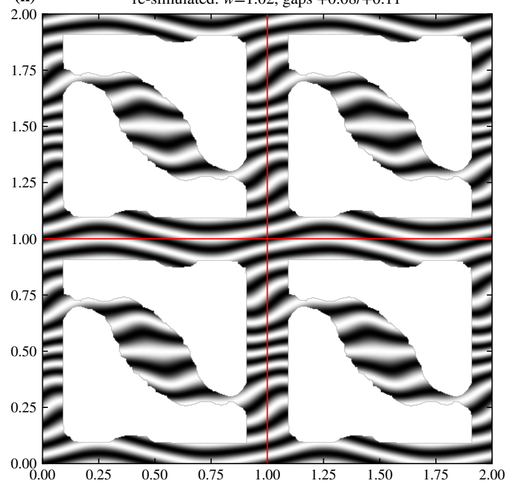
(i) as-manufactured tows, pitch d=0.167
re-simulated: w=0.84, gaps -0.14/+0.36



(j) as-manufactured tows, pitch d=0.111
re-simulated: w=0.65, gaps +0.21/+0.52



(k) as-manufactured tows, pitch d=0.083
re-simulated: w=1.02, gaps +0.08/+0.11



(l) as-manufactured tows, pitch d=0.083
re-simulated: w=0.55, gaps +0.59/+0.57

